

Mark Christof

CS 499 Computer Science Capstone

The artifact selected for Enhancement One is the Python CRUD module originally developed for the CS 340 Animal Shelter project. This artifact implements a reusable Create, Read, Update, and Delete interface for interacting with a MongoDB database and serves as the backend data access layer for the application. It was selected because it reflects core software design and engineering principles, including modular structure, clear method responsibilities, and defensive programming practices, making it well-suited for refinement and inclusion in a professional ePortfolio.

For this enhancement, I focused on improving the clarity and maintainability of the existing code without altering its functionality. Based on feedback received during the code review milestone, I added targeted line-level comments in areas where the intent of specific statements may not be immediately obvious to a reviewer. These additions complement the existing block-level comments and docstrings by providing clearer explanations of design decisions at key points in the code. In particular, I clarified how default query behavior is handled, why certain database fields are excluded from returned results, and how method parameters influence update behavior. Minor formatting adjustments were also made to ensure the file reads cleanly and consistently.

These changes strengthen the overall software design of the artifact by improving communication and readability while preserving the original behavior of the system. Enhancing the code in this way supports long-term maintainability and makes the module easier for other developers or reviewers to understand and evaluate. This enhancement demonstrates professional software engineering practices by emphasizing clear documentation, thoughtful design decisions,

and attention to detail, all of which are essential qualities for backend development and for presenting work in a public ePortfolio.